Optimal Time-Stepping with Non-Iterative Defect Correction for Coupled Reaction–Diffusion Equations

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Repository: `OTS-NIDC` (this paper is generated from it)

1 Review for Single Variable PDEs

Throughout, $\delta_x^2 u_j = (u_{j+1} - 2u_j + u_{j-1})/\Delta x^2$ is the centered second difference and refinement means $\Delta x \to 0$ with Δt slaved to Δx.

1.1 The scalar family and the optimal time step

Consider the deterministic family

$$u_t = D\, u_{xx} + \alpha\, u, \qquad D > 0,\ \alpha \in \mathbb{R}, \tag{1}$$

whose stochastic motivation (the moment equations of the linear SPDE of the companion Part II, with $\alpha = \frac{k(k-1)}{2}\sigma^2$) is deferred there. The pure diffusion case $\alpha = 0$ is exactly Chu's setting (Chu, 2011).

Apply forward Euler in time and the three-point stencil in space. Taylor expansion of both sides, followed by repeated differentiation of (1) to convert time derivatives of u into space derivatives (the operators $D\partial_{xx}$ and α commute, giving the binomial identity $u_{t^k} = (D\partial_{xx} + \alpha)^k u$), yields the local truncation error organized by derivative type:

$$\mathcal{T} \;=\; \underbrace{\left(\tfrac{D^2\Delta t}{2} - \tfrac{D\Delta x^2}{12}\right) u_{xxxx}}_{\text{Group 1}} \;+\; \underbrace{\alpha D\, \Delta t\, u_{xx}}_{\text{Group 2}} \;+\; \underbrace{\tfrac{\alpha^2 \Delta t}{2}\, u}_{\text{Group 3}} \;+\; O(\Delta t^2, \Delta x^4). \tag{2}$$

Only Group 1 couples Δt to Δx. Setting it to zero:

Theorem 1 (Optimal time step (Chu, 2011))**.** *For (1) discretized by forward Euler / centered differences, the unique step size annihilating the leading (u_{xxxx}) truncation term is*

$$\boxed{\Delta t_{\text{opt}} = \frac{\Delta x^2}{6D}} \tag{3}$$

and no single step annihilates both the u_{xxxx} and u_{xxxxxx} truncation terms, since the level-2 condition is $\Delta t = \Delta x^2/(\sqrt{60}\, D) \neq \Delta t_{\text{opt}}$.

(Proof: one-line cancellation in (2); uniqueness by the level-2 computation. Full algebra in `sign-change.tex` §1.)

Proposition 2 (α-independence)**.** Δt_{opt} *is independent of the reaction parameter* α*: the cancellation operates on the diffusive group alone. For the stochastic hierarchy this makes* Δt_{opt} *independent of the noise amplitude* σ *and of the moment order* k*.*

Following the derivation inspecting the truncation error, at $\Delta t = \Delta t_{\text{opt}}$ the Group-1 term vanishes and the surviving residual is

$$\mathcal{T}\big|_{\Delta t_{\text{opt}}} \;=\; \frac{\alpha \Delta x^2}{6}\, u_{xx} + \frac{\alpha^2 \Delta x^2}{12 D}\, u \;+\; O(\Delta x^4). \tag{4}$$

Remark 3 (Sign convention)**.** The truncation error $\mathcal{T}$ as assembled in (2) is the *forward* residual: exact $-$ scheme $= \Delta t\, \mathcal{T} + \cdots$. Cancelling it therefore requires *adding* $\Delta t\, \mathcal{T}$; all corrections below carry this sign.

Lemma 4 (Closed form correction term)**.** *The corrected factor at any given time step is,*

$$C_j^{(1)} = +\frac{\alpha\, \Delta x^4}{36 D}\, \delta_x^2 u_j^n \;+\; \frac{\alpha^2\, \Delta x^4}{72 D^2}\, u_j^n. \tag{5}$$

Theorem 5 (First-level accuracy)**.** *The corrected scheme*

$$u_j^{n+1} = u_j^n + \Delta t_{\text{opt}} \Big[D\, \delta_x^2 u_j^n + \alpha\, u_j^n \Big] + C_j^{(1)} \tag{6}$$

has step error $O(\Delta x^6)$ *and global order* 4*: the correction* $C^{(1)}$ *is itself* $O(\Delta x^4)$*, so its evaluation with the second-order operator* δ_x^2 *contributes only* $O(\Delta x^6)$ *to the step — the three-point stencil is mathematically sufficient at* $p = 1$*.*

Status: [V]. Measured orders and errors, with the harness described in Appendix B.2:

Table 1: Scalar $p = 1$ convergence, $u_t = D u_{xx} + \alpha u$ with $D = 0.1$, $\alpha\sigma^2 = 0.25$, Neumann BCs, $T = 0.05$; L^2 error against the exact cosine-series solution. Data: `code/implementation/results.md` (generated by `convergence_test.py`, 2026-07). All methods use the same grid and step size Δt_{opt} at each level.

Method	$N_x = 32$	$N_x = 64$	$N_x = 128$	$N_x = 256$
OTS-NIDC ($p = 1$)	8.54e-09	5.26e-10	3.27e-11	2.04e-12
Forward Euler	1.23e-05	3.02e-06	7.52e-07	1.88e-07
4th-order FD + FE	1.52e-05	3.79e-06	9.45e-07	2.36e-07
Reference expm	2.77e-05	6.81e-06	1.70e-06	4.23e-07
Measured order (log–log slope, mean over levels):				
OTS-NIDC	4.01	(FE / 4th-order FD / expm: 2.00–2.01)		

The 4th-order-FD row carries the paper's cleanest negative control: high-order spatial accuracy alone buys nothing when the step size is slaved to Δx^2 — the $O(\Delta t)$ time error caps the whole scheme at second order. The boost must come from the time-step structure; that is the method's thesis, verified.

1.2 Stability of Single Variable Systems

Set $u_j^n = G^n e^{i\kappa j\Delta x}$; the three-point Laplacian has symbol $\delta_x^2 u_j^n \to -\frac{4}{\Delta x^2} s\, u_j^n$ with $s = \sin^2(\kappa\Delta x/2) \in [0,1]$. Measure the step against the optimum, $\tau := \Delta t_{\text{opt}} = \Delta x^2/(6D)$:

$$r = \frac{\Delta t}{\tau} = \frac{6D\Delta t}{\Delta x^2}, \qquad \eta = \frac{\alpha\Delta x^2}{D}, \qquad a = \alpha\Delta t = \frac{r\eta}{6},$$

where r is the step ratio used throughout §3–4 (there the forced step is $h = r\tau$ with mismatch $\varepsilon = (r-1)\tau$), η is the step-independent reaction strength, and the legacy diffusion number is $\mu = r/6$. The exact semi-discrete one-step factor is $E(s) = e^{-\frac{2r}{3}s+a}$: since the reaction grows, $|G| \le 1$ is the wrong requirement, and we use the von Neumann condition with growth allowance

$$|G(s)| \le 1 + C\,\Delta t \quad \text{uniformly in } s \in [0,1], \tag{7}$$

along refinement paths $\Delta t = r\tau$ with r fixed (Lax–Richtmyer); the sharper natural comparison is $|G(s)| \le e^a$.

We proceed in four steps: the amplification factors themselves (Lemma 6); stability at the optimal step, which is what makes Δt_{opt} safe for every moment of the hierarchy (Lemma 7); the grid-scale ceilings away from the optimum, which fix the operating ranges of §3–4 (Lemma 8); and the scope of the frozen correction (Remark 9). Theorem 11 collects the result.

Lemma 6 (Amplification factors)**.** *With $r = \Delta t/\tau$ and $\eta = \alpha\Delta x^2/D$,*

$$G_{\text{FE}} = 1 - \tfrac{2r}{3}s + \tfrac{r\eta}{6}, \tag{8}$$

$$G_{\text{NIDC}} = 1 - \tfrac{2r}{3}s + \tfrac{r\eta}{6} - \tfrac{\eta}{9}s + \tfrac{\eta^2}{72}, \tag{9}$$

and $G_{\text{NIDC}} = G_{\text{FE}} - \frac{\eta}{9}s + \frac{\eta^2}{72}$ is strictly decreasing in s on $[0,1]$.

Proof. Substituting the symbol of δ_x^2 into the uncorrected update gives (8). For the corrected scheme, the $C^{(1)}$ terms of (5) contribute $\frac{\alpha\Delta x^4}{36D}\cdot\left(-\frac{4s}{\Delta x^2}\right) = -\frac{\eta}{9}s$ and $\frac{\alpha^2\Delta x^4}{72D^2} = \frac{\eta^2}{72}$, giving (9) and the perturbation identity. The slope in s is $-\left(\frac{2r}{3} + \frac{\eta}{9}\right) < 0$. □

Both perturbation terms are $O(\Delta x^2)$, so the extremes over $s \in [0,1]$ sit at the endpoints $s = 0$ (physical mode) and $s = 1$ (grid mode); every bound below is therefore an endpoint check.

Lemma 7 (Stability at the optimal step)**.** *At $r = 1$ (so $a = \eta/6$),*

$$G_{\text{NIDC}}(s) = \left(1 - \tfrac{2}{3}s\right)(1+a) + \tfrac{a^2}{2}, \tag{10}$$

and for every $\eta \ge 0$ and all $s \in [0,1]$,

$$\tfrac{1}{3}(1+a) + \tfrac{a^2}{2} \;\le\; G_{\text{NIDC}}(s) \;\le\; 1 + a + \tfrac{a^2}{2} \;\le\; e^a \qquad (a \ge 0).$$

In particular $G_{\text{NIDC}} > 0$ (no oscillatory modes) and (7) holds with no restriction on η.

Proof. Substitute $r = 1$ in (9); $\eta = 6a$ gives (10). Since $1 - \frac{2}{3}s \in [\frac{1}{3}, 1]$, the bracket bounds follow, and $1 + a + \frac{a^2}{2} \le e^a$ is the tail of the exponential series. The lower end is positive, giving the two-sided bound. □

This is the headline safety fact: the correction never destabilizes the optimal step, at any reaction strength. The offset $+\eta^2/72$ is not destabilizing — it is what makes $G(0)$ match e^a to $O(a^2)$. Plain forward Euler at the same r has $G_{\text{FE}}(s) = (1 - \frac{2}{3}s) + a$, inside $[\frac{1}{3} + a,\ 1 + a]$.

Lemma 8 (Grid-scale ceilings for general r)**.** *For $0 < \eta < 8$, requiring the lower side of (7) at $s = 1$, i.e. $G(1) \geq -(1 + a)$, gives*

$$r_{\mathrm{FE,max}} = \frac{6}{2-\eta} = \frac{3}{1-\eta/2}, \tag{11}$$

$$r_{\mathrm{NIDC,max}} = \frac{3(2-\eta/9+\eta^2/72)}{2-\eta}, \tag{12}$$

with ratio

$$\frac{r_{\mathrm{NIDC,max}}}{r_{\mathrm{FE,max}}} = 1 - \frac{\eta}{18} + \frac{\eta^2}{144}. \tag{13}$$

Proof. At $s = 1$ the factors are affine in r: $G_{\mathrm{FE}}(1) = 1 - \frac{2r}{3} + \frac{r\eta}{6}$ and $G_{\mathrm{NIDC}}(1) = G_{\mathrm{FE}}(1) - \frac{\eta}{9} + \frac{\eta^2}{72}$. Setting each equal to $-(1+a)$ with $a = \frac{r\eta}{6}$ and solving the linear equation in r gives (11)–(12); the ratio (13) is their quotient. □

Two directions matter and should not be conflated. The *shrink* is relative to forward Euler at the same η: -1.6% at the coarse $\eta = 0.3$ test point ($r_{\mathrm{FE,max}} = 3.529$, $r_{\mathrm{NIDC,max}} = 3.473$, ratio 0.98396), and $O(\Delta x^2)$. The *absolute* ceiling $\Delta t_{\max} = r_{\mathrm{NIDC,max}}\tau = \frac{2-\eta/9+\eta^2/72}{2-\eta} \cdot \frac{\Delta x^2}{2D}$ loosens with η relative to the $\eta = 0$ ceiling $\Delta x^2/(2D)$ ($+16\%$ at $\eta = 0.3$). (The stricter test $G(1) \geq -1$ gives ceilings $6/(4-\eta)$ and $6(2-\eta/9+\eta^2/72)/(4-\eta)$, with the same ratio.) These ceilings are exactly what §3 uses as its operating range ($r < 4$) and §4 as its switching rule (r_c).

Remark 9 (Scope of the frozen correction)**.** $C^{(1)}$ is derived at $\Delta t = \Delta t_{\mathrm{opt}}$ and is independent of Δt. At $s = 0$,

$$G_{\mathrm{NIDC}}(0) - e^a \;\approx\; \frac{\eta^2}{72}\big(1 - r^2\big),$$

which is ≤ 0 iff $r \geq 1$. For $r < 1$ the excess is Δt-independent, so at fixed Δx with $\Delta t \to 0$ the accumulated growth $(1 + \mathrm{const})^{T/\Delta t}$ diverges. Stability statements therefore refer to refinement paths with r fixed, where the excess is $O(\Delta t^2)$ and (7) holds; the $\Delta t/\Delta t_{\mathrm{opt}} \in \{0.5, \ldots, 2\}$ table is a sweep over such paths, not a fixed-Δx study. The method is designed for $r = 1$.

Corollary 10 (Pair reduction)**.** *For a pair of species with $D \to D_i + D_j$ and $\alpha \to \alpha_{ij} = R_{ij}$, (10) holds verbatim with $r_{ij} = 1$ at $\Delta t_{\mathrm{opt}}^{(ij)}$.*

Theorem 11 (Von Neumann stability)**.** *With $s = \sin^2(\kappa\Delta x/2)$, $r = \Delta t/\tau$, $\eta = \alpha\Delta x^2/D$, the factors are (8)–(9). At the optimal step $r = 1$ the corrected scheme is unconditionally stable for every $\eta \geq 0$ (Lemma 7); for general r and $0 < \eta < 8$ the grid-scale bound tightens from (11) to (12), a relative shrink of $1 - \eta/18 + O(\eta^2)$, i.e. $O(\Delta x^2)$, about -1.6% at $\eta = 0.3$. The optimal step $r = 1$ sits a factor of at least 3 inside the CFL ceiling for both schemes.*

Proof. Assemble Lemmas 6, 7 and 8; the endpoint reduction follows from the monotonicity in Lemma 6. □

Status: [V](analytic derivation above; numerical spot-checks against the harness of Appendix B.2; the earlier "expands the CFL" conclusion was an artifact of the pre-correction sign and has been fixed in the same release). The correction is stability-neutral to first order; it never threatens the optimal step. Note the honest asymmetry: this is a *scalar* analysis; the coupled discrete schemes of §2 have no independent stability proof (open item C).

1.3 Efficiency profile

The method's cost argument is architectural, not just arithmetic: versus multi-stage Runge–Kutta, OTS-NIDC is single-pass (one read, one write of the state vector; no stage storage, no stage barriers); versus compact schemes, it is strictly local (no tridiagonal solves), so it maps onto GPUs and bandwidth-bound hardware without global communication. At $N_x = 256$ the verified scalar runs show $9.2 \times 10^4\times$ lower error than forward Euler at identical grid and step ($1.4 \times 10^3\times$ already at $N_x = 32$) [V]— order wins over constants at refinement.

2 The Step to Coupled PDEs

We now couple species to allow for multi-linear systems, a direct expansion from the canon work we expand on!

2.1 Reaction-Diffusion Family and OTS

For N species with diagonal diffusion, linear reaction coupling, and (here) zero noise — the deterministic skeleton of the moment systems to come —

$$\partial_t u^{(i)} = D_i\, \partial_{xx} u^{(i)} + \sum_j R_{ij} u^{(j)}. \tag{14}$$

Lemma 12 (OTS-NIDC for fixed indices)**.** *For equation (14), the optimal step for each $u^{(i)}$ individually, as derived in the single variable case, is exactly $\Delta t^{(i)}_{\text{opt}} = \frac{\Delta x^2}{6D_i}$, with first-level corrections $C^{(1),(i)}$ of the same algebraic form as (5), derived term-by-term in `OTS-NIDC.md` §8–9.*

With this, we get the desired theorem as follows.

Theorem 13 (First-level accuracy for fixed indices)**.** *The scheme*

$$u_j^{(i),n+1} = u_j^{(i),n} + \Delta t^{(i)}_{\text{opt}} \left[D_i \delta_x^2 u_j^{(i),n} + \sum_\ell R_{i\ell} u_j^{(\ell),n} \right] + C^{(1),(i)} \tag{15}$$

has similar step error and global order as given by Theorem 5.

2.1.1 Competing Time Steps and Subcycling

Notice that the scheme presented by Theorem 13 relies on having $u_j^{(\ell),n}$, the value of the separate PDE $u^{(\ell)}$ at time-step n. However, since each equation carries its own optimal step, the required value is not readily available whenever the diffusivities differ, $D_i \neq D_j$ for some $i \neq j$, i.e. whenever no single Δt_{opt} serves the whole system.

The two failure modes of a naive single step frame the problem.

Failure 1: shared step without correction. If species 1 (fast, D_1, optimal step $\tau_1 = \Delta t^{(1)}_{\text{opt}}$) is forced onto species 2's optimum $h = \tau_2 = \Delta t^{(2)}_{\text{opt}}$, its Group-1 cancellation breaks. The one-step identity

$$e(h) \;=\; hD_1\, \tilde{u}_{xxxx} \left[\tfrac{hD_1}{2} - \tfrac{\Delta x^2}{12} \right] + O(h^3) \tag{16}$$

is exact for every h — where $\tilde{u}_{xxxx} = \partial_x^4 u^{(1)}$ is evaluated on the exact solution at the step, the representative of the LTE remainder as in §1 — and writing $h = \tau_1 + \varepsilon$ turns the bracket into $\varepsilon D_1/2$ *identically* (no Taylor truncation in ε).

We remain, however, that this stencil is not ideal for each step, and it leads to questions of stability. In fact,

Failure 2: naive splitting. Lie–Trotter splitting with the coupling frozen over a macro step H incurs a sustained $O(H)$ forcing error; accumulated over T/H cycles this is a global $O(H) = O(\Delta x^2)$ error — worse than the uncorrected base scheme. Freezing is not viable; coupling data must be interpolated.

The rest of this paper introduces methods of attempting to corroborate the differences between these time-steps in ways to recover the accuracy that we would lose by just picking one or the other.

3 Regime 1: Explicit Correction Single Stepping

Recall the mismatch identity (16): forcing the fast equation (optimal step $\tau_1 = \Delta t_{\text{opt}}^{(1)} = \Delta x^2/(6D_1)$) onto a larger step h re-involves the u_{xxxx} term, and writing $h = \tau_1 + \varepsilon$ reduces its Group-1 coefficient to exactly $\frac{D_1^2 \varepsilon}{2}$ per unit time (the per-step error carries the extra factor h). With this, we can formulate a first method of attempting to correspond any two coupled equations.

For any two equations with distinct $\Delta t_{\text{opt}}^{(1)} < \Delta t_{\text{opt}}^{(2)}$, we evolve both equations by the shared step $h = \Delta t_{\text{opt}}^{(2)}$ such that $u^{(2)}$ is optimally solved by nature, and the discretization of $u^{(1)}$ is evolved with the forced step $h = \tau_1 + \varepsilon$ such that the explicit mismatch correction term recovers order 4 accuracy. Namely,

Lemma 14. *Forcing the time-step h on the equation for species i, whose optimal step is $\tau_i := \Delta t_{\text{opt}}^{(i)} = \Delta x^2/(6D_i)$, write $h = \tau_i + \varepsilon$. The* ***explicit mismatch correction term*** *is*

$$C_{mismatch}^{(i)} = \frac{h D_i^2 \varepsilon}{2} \delta_x^4 u^{(i),n}$$

where $\delta_x^4 = (\delta_x^2)^2$ is evaluated via a five-point stencil and $u^{(i),n}$ is the i-th species evaluated at time n.

Then,

Theorem 15 (OTS-NIDC accuracy fixed indices with time-step forcing)**.** *For equation $u^{(i)}$ with forced time-step h, the scheme*

$$u_j^{(i),n+1} = u_j^{(i),n} + h \left[D_i \delta_x^2 u_j^{(i),n} + \sum_{\ell} R_{i\ell} u_j^{(\ell),n} \right] + C^{(1),(i)} + C_{mismatch}^{(i)} \tag{17}$$

has similar step error and global order as given by Theorem 5.

3.1 Stability

As we've established that the regime for individual equations remains the same, we prove stability by showing the correction factor described in Lemma 14 is similarly a perturbation on the simple FE scheme. We follow the scalar analysis of §1, now allowing the actual step h to differ from the

optimal step $\tau = \Delta t_{\text{opt}} = \frac{\Delta x^2}{6D}$ (here $D = D_1$, the fast species). Write $h = \tau + \varepsilon$ and $r = \frac{h}{\tau}$; this is the r of §1, with $\mu = r/6$ and $a = \alpha h = r\eta/6$. We first establish stability of the mismatch-corrected diffusion step, then restore the reaction terms.

Lemma 16 (Mismatch correction and amplification factor)**.** *For $u_t = Du_{xx}$, the corrected update is $u^{n+1} = u^n + hD\delta_x^2 u^n + \frac{hD^2\varepsilon}{2}\delta_x^4 u^n$, where $\delta_x^4 = (\delta_x^2)^2$. With $s = \sin^2\left(\frac{\kappa\Delta x}{2}\right) \in [0,1]$, the diffusion-only amplification factor of this update is $G_{\text{mis}}(s) = 1 - \frac{2r}{3}s + \frac{2r(r-1)}{9}s^2$.*

Proof. By Lemma 14, cancellation requires adding $\frac{hD^2\varepsilon}{2}\delta_x^4 u^n$.

The symbols of δ_x^2 and δ_x^4 are $-\frac{4s}{\Delta x^2}$ and $\frac{16s^2}{\Delta x^4}$. Since $\frac{hD}{\Delta x^2} = \frac{r}{6}$ and $\frac{hD^2\varepsilon}{\Delta x^4} = \frac{r(r-1)}{36}$ (substituting $h = r\tau$, $\varepsilon = (r-1)\tau$, $\tau = \Delta x^2/6D$), the update factor is $1 + hD\cdot\left(-\frac{4s}{\Delta x^2}\right) + \frac{hD^2\varepsilon}{2}\cdot\frac{16s^2}{\Delta x^4} = G_{\text{mis}}(s) = 1 - \frac{2r}{3}s + \frac{2r(r-1)}{9}s^2$. At $r = 1$, the mismatch vanishes and this reduces to the diffusion part of the optimal-step factor (10) from §1. □

Unlike the reaction corrections in §1, the mismatch contribution need not vanish under refinement at fixed grid frequency: its symbol is $\frac{2r(r-1)}{9}s^2$. We therefore bound the complete factor rather than assume that the correction is uniformly small.

Lemma 17 (Diffusive stability interval)**.** *On an infinite or periodic uniform grid, the mismatch-corrected diffusion step satisfies $|G_{\text{mis}}(s)| \le 1$ for every $s \in [0,1]$ whenever $0 < r \le 4$.*

Proof. The upper bound follows from $G_{\text{mis}}(s) - 1 = \frac{2rs}{9}\left((r-1)s - 3\right) \le 0$: for $r \le 1$ the second factor is nonpositive; for $1 < r \le 4$ it is bounded by $(r-1)s \le r - 1 \le 3$.

For the lower bound, completing the square gives $G_{\text{mis}}(s) = \frac{2}{9}\left(rs - \frac{s+3}{2}\right)^2 + 1 - \frac{(s+3)^2}{18}$, and the residual term satisfies $1 - \frac{(s+3)^2}{18} \ge \frac{1}{9}$ because $(s+3)^2 \le 16$ for $0 \le s \le 1$. Hence $\frac{1}{9} \le G_{\text{mis}}(s) \le 1$; the constant $\frac{1}{9}$ is sharp, attained at $(r,s) = (2,1)$.

For $r > 4$, the grid-scale factor is $G_{\text{mis}}(1) = 1 + \frac{2r(r-4)}{9} > 1$. Thus $r = 4$ is the sharp boundary for this Fourier criterion. □

In particular, the correction preserves diffusive stability throughout the mismatch-corrected range $0 < r \le 4$ — extending the uncorrected forward Euler ceiling $r \le 3$ — and in particular for shortened remainder steps $0 < r < 1$. We retain $r < 4$ as the operating restriction (the strict form needed below) and use it to recover the reaction-growth bound from §1.

For $u_t = Du_{xx} + \alpha u$, the reaction correction must also be evaluated at the actual step. The same Taylor expansion used in §1 gives $C_{\text{react}}u^n = h^2\alpha D\delta_x^2 u^n + \frac{h^2\alpha^2}{2}u^n$. At $h = \tau$, this is exactly the correction previously derived. Together with the mismatch term, it cancels the leading residual at a general step h.

Remark 18 (Relation to the factors of §1)**.** G_{mis} is the *diffusion-only* building block and is not directly comparable to G_{FE}, which includes the reaction. The full corrected factor derived in Theorem 19 below is named $G_{\text{NIDC},h}$: it is the forced-step analogue of G_{NIDC}, and plain forward Euler at the same step has $G_{\text{FE}}(s) = \left(1 - \frac{2r}{3}s\right) + a$, so

$$G_{\text{NIDC},h}(s) - G_{\text{FE}}(s) \;=\; \underbrace{\frac{2r(r-1)}{9}s^2}_{\text{mismatch}} \qquad \underbrace{-\frac{2ra}{3}s \;+\; \frac{a^2}{2}}_{\text{reaction corrections at step } h} \quad , \qquad a = \frac{r\eta}{6},$$

which reduces at $r = 1$ (where $a = \eta/6$) to exactly the perturbation $-\frac{\eta}{9}s + \frac{\eta^2}{72}$ of §1. Off the optimum the two conventions differ: §1 evaluates the coefficients of $C^{(1)}$ (derived at $r = 1$) at a

general step, while the present section rescales the reaction corrections to h and adds the mismatch term; the two coincide at $r = 1$ and differ at $O(r-1)$ elsewhere (see the notation dictionary, Appendix B).

Theorem 19 (Scalar stability with mismatch correction)**.** *Let $\alpha \geq 0$, and consider $u^{n+1} = u^n + h(D\delta_x^2 u^n + \alpha u^n) + \frac{hD^2(h-\tau)}{2}\delta_x^4 u^n + h^2\alpha D\delta_x^2 u^n + \frac{h^2\alpha^2}{2}u^n$. On an infinite or periodic uniform grid, if $0 < h < 4\tau$, then its amplification factor satisfies $|G_{\mathrm{NIDC},h}(s)| \leq e^{\alpha h}$ uniformly in $s \in [0,1]$.*

Consequently, any sequence of admitted steps, including optimal substeps and shortened remainder steps, satisfies $\|u^{n_f}\|_{\ell^2} \leq e^{\alpha T}\|u^0\|_{\ell^2}$, where $T = \sum_{n=0}^{n_f-1} h_n$ and n_f is the final step index.

Proof. Set $a = \alpha h$ and $r = \frac{h}{\tau}$. Substituting the Fourier symbols gives $G_{\mathrm{NIDC},h}(s) = G_{\mathrm{mis}}(s) + a\left(1 - \frac{2r}{3}s\right) + \frac{a^2}{2}$. The preceding lemma gives $\frac{1}{9} \leq G_{\mathrm{mis}}(s) \leq 1$. Moreover, for $0 < r < 4$, $-\frac{5}{3} < 1 - \frac{2r}{3}s \leq 1$. Therefore, for the upper side, $G_{\mathrm{NIDC},h}(s) \leq 1 + a + \frac{a^2}{2} \leq e^a$, and for the lower side, $G_{\mathrm{NIDC},h}(s) \geq \frac{1}{9} - \frac{5a}{3} + \frac{a^2}{2} > -(1+a) \geq -e^a$, where the middle inequality is $\frac{10}{9} - \frac{2a}{3} + \frac{a^2}{2} > 0$, which holds for every $a \geq 0$. Hence $|G_{\mathrm{NIDC},h}(s)| \leq e^a$.

Over successive steps, the amplification factors multiply. Hence their product has magnitude at most $\prod_{n=0}^{n_f-1} e^{\alpha h_n} = e^{\alpha T}$. The discrete ℓ^2 estimate follows from Parseval's identity. □

Thus the mismatch correction preserves the scalar growth allowance used in §1 throughout the admitted range $0 < r < 4$. In a subcycle, full steps have $r = 1$ and any shortened remainder has $0 < r < 1$, so both satisfy the same estimate. This establishes the scalar building block; stability of off-diagonal reaction coupling and its interpolation still requires analysis of the complete coupled update.

3.2 Advantage

4 Regime 2: Interpolation and Subcycling

4.1 Setup and switching rule

Two species with $\Delta t_{\mathrm{opt}}^{(1)} < \Delta t_{\mathrm{opt}}^{(2)}$, so $D_1 > D_2$ and species 1 is the fast one. Define the step ratio $r = \frac{\Delta t_{\mathrm{opt}}^{(2)}}{\Delta t_{\mathrm{opt}}^{(1)}} = \frac{D_1}{D_2} > 1$. This is the same $r = h/\tau$ as in Regime 1, with $h = \Delta t_{\mathrm{opt}}^{(2)}$ forced onto the fast equation and $\tau = \Delta t_{\mathrm{opt}}^{(1)}$. The mismatch is $\varepsilon = \Delta t_{\mathrm{opt}}^{(2)} - \Delta t_{\mathrm{opt}}^{(1)} = (r-1)\,\Delta t_{\mathrm{opt}}^{(1)}$, as in Regime 1.

- Macro step $H = \Delta t_{\mathrm{opt}}^{(2)}$. Write $n_r = \lfloor r \rfloor$, $\theta = r - n_r \in [0,1)$, and remainder $\rho = H - n_r\,\Delta t_{\mathrm{opt}}^{(1)} = \theta\,\Delta t_{\mathrm{opt}}^{(1)} \in [0, \Delta t_{\mathrm{opt}}^{(1)})$.
- Switching rule: Regime 1 is stable for $r < 4$ (Theorem 19), so Regime 2 is needed for $r \geq r_c$ with $r_c \in [3,4)$ a design parameter.
  - $r_c = 3$ is the conservative choice (the uncorrected forward Euler ceiling).
  - Larger r_c is supported by Fourier stability up to $r = 4$, but at $r = 4$ the grid mode is only marginally stable ($G_{\mathrm{mis}}(1) = 1$) and the mismatch term is $O(1)$ at grid scale. Say this explicitly and justify the choice of r_c numerically.
- Ordering within a macro step (slow first):
  1. Advance $u^{(2)}$ from t^k to $t^{k+1} = t^k + H$ with its own optimal step, using $u^{(1)}$ at t^k only (explicit coupling).

2. Sub-cycle $u^{(1)}$ over $[t^k, t^{k+1}]$, reading $u^{(2)}$ from an interpolant built on the two known levels (and optionally history).

This makes the slow-species coupling data a true interpolation, never an extrapolation.

- Only the *coupling data* needs interpolating. The fast species' own state is advanced by the Regime 1 scheme.

4.2 How much interpolation is required

Let $I_q u^{(2)}$ denote the Lagrange interpolant in time of degree q through $q+1$ known slow levels, and let p be the level of the correction tower (global order $2p+2$).

Lemma 20 (Interpolation error)**.** *For $t \in [t^k, t^{k+1}]$ and $u^{(2)}$ sufficiently smooth in time,*

$$\left|u^{(2)}(t) - (I_q u^{(2)})(t)\right| \le C_q \left(\Delta t^{(2)}_{\text{opt}}\right)^{q+1} \max |\partial_t^{q+1} u^{(2)}|,$$

where C_q is the Lagrange node-polynomial constant. In particular $C_1 = 1/8$.

- Proof: standard Lagrange remainder, no new content. Cite and state the constants for $q = 1, 2$.
- Remark: since $\partial_t^2 u = D_2^2 \partial_x^4 u + \dots$ and $\Delta t^{(2)}_{\text{opt}} = \Delta x^2/(6D_2)$, the linear case gives an error of about $\Delta x^4 |u_{xxxx}|/288$, independent of D_2.

Theorem 21 (Degree of interpolation)**.** *Suppose the fast update uses coupling data $R_{12}(I_q u^{(2)})$. Each sub-step has local error $O(\Delta x^{2q+4})$ from the coupling data, hence the global order contributed is $\min(2p+2,\, 2q+2)$. In particular, $q \ge p$ is necessary and sufficient to preserve order $2p+2$.*

- Proof sketch to write out: the interpolation error enters multiplied by the step $h \sim \Delta x^2$, then accumulate over $O(\Delta x^{-2})$ steps. Constants grow with r (the interpolation interval is $\Delta t^{(2)}_{\text{opt}} = r\, \Delta t^{(1)}_{\text{opt}}$) but r is bounded by design.
- Consequence table for the paper:
  - $p = 1$: linear interpolation (2 nodes: t^k, t^{k+1}).
  - $p = 2$: quadratic (3 nodes, adds t^{k-1}).
  - Variant: cubic Hermite dense output using the stored derivatives $D_2 \delta_x^2 u^{(2)} + Ru$ that the slow step already computes. Needs no history, gives higher-order coupling error than $p = 1$ requires.
- **[Open derivation]** The correction $C^{(1),(1)}$ on each substep includes a cross term containing $R_{12}\, \partial_t u^{(2)}$. Decide how $\partial_t u^{(2)}$ is supplied (finite difference of the two slow levels, or the Hermite derivative) and check it does not lower the order.

4.3 Closing the macro step

After n_r full steps of $\Delta t^{(1)}_{\text{opt}}$, the fast species has advanced $n_r\, \Delta t^{(1)}_{\text{opt}}$ and still needs $\rho = \theta\, \Delta t^{(1)}_{\text{opt}}$ to reach t^{k+1}.

4.3.1 Method A: remainder step

- Take one final step of size ρ with mismatch parameter $\varepsilon_\rho = \rho - \Delta t^{(1)}_{\text{opt}} = (\theta - 1)\Delta t^{(1)}_{\text{opt}} < 0$ (the Regime 1 ε for this step) in Lemma 14, coupling data from $I_q u^{(2)}$ at the beginning of the step.
- Uses a five-point stencil for δ_x^4; needs no solution history.

Theorem 22 (Subcycled scheme with remainder step)**.** *Let $r \geq r_c$, $q \geq p$. The scheme above has global order $2p + 2$ (order 4 at $p = 1$).*

- Proof outline: Theorem 21 for the coupling data; Theorem 15 for each step (the remainder has step ratio $\rho/\Delta t^{(1)}_{\text{opt}} = \theta < 1$); telescoping over the $n_r + 1$ steps per macro step.
- Stability: scalar building block is Theorem 19, since full steps have step ratio $h/\tau = 1$ and the remainder has $0 < h/\tau < 1$. Coupled stability still open (same status as Regime 1).
- Failure mode to remark on: $\theta \to 0$ gives a vanishingly short remainder. Fine for stability ($G_{\text{mis}} \to 1$), but check that the mismatch term $\varepsilon_\rho \to -\Delta t^{(1)}_{\text{opt}}$ behaves.

4.3.2 Method B: extrapolated point

- Take only the n full optimal steps, with no mismatch correction. Then advance the *state* to t^{k+1} by extrapolating the last $q + 1$ fast levels forward by a fraction θ of a step.
- The extrapolation error lands on the state directly rather than being multiplied by h, so the degree requirement is one higher than for interpolation.

Lemma 23 (Extrapolation error)**.** *Let E_q be the degree-q extrapolant through $q+1$ consecutive fast levels spaced $\Delta t^{(1)}_{\text{opt}}$ apart, evaluated at $\theta\,\Delta t^{(1)}_{\text{opt}}$ beyond the last node. Then*

$$\left|u^{(1)} - E_q u^{(1)}\right| \leq \Lambda_q(\theta)\left(\Delta t^{(1)}_{\text{opt}}\right)^{q+1} \max|\partial_t^{q+1} u^{(1)}|, \qquad \Lambda_q(\theta) = \frac{\prod_{i=0}^{q}(\theta + i)}{(q+1)!}.$$

- $\Lambda_1(\theta) = \theta(1+\theta)/2$ and $\Lambda_2(\theta) = \theta(\theta+1)(\theta+2)/6$; both are ≤ 1 for $\theta \in [0, 1]$.
- Remark: since $\partial_t^{q+1} u \sim D_1^{q+1} \partial_x^{2q+2} u$ and $\Delta t^{(1)}_{\text{opt}} = \Delta x^2/(6D_1)$, the constant is independent of D_1.
- Accuracy range: the extrapolation distance is $\theta < 1$ by construction, i.e. less than one optimal step beyond the last node. Λ_q grows factorially for longer distances, so this should be stated as the reason the method is limited to a single sub-step of extrapolation.

Theorem 24 (Degree of extrapolation)**.** *The extrapolated point contributes a local error $O(\Delta x^{2q+2})$ per macro step, hence global order $\min(2p + 2, 2q)$. Order $2p + 2$ therefore requires $q \geq p + 1$: at $p = 1$ this means quadratic extrapolation from three levels, and linear extrapolation caps the scheme at order 2.*

- Proof outline: Lemma 23 gives the local error. Telescoping over $T/H = O(\Delta x^{-2})$ macro steps.

- Remark: if the extrapolated value were only used as coupling data (rather than overwriting the state), the error would again be multiplied by h and the interpolation rule $q \geq p$ would apply. Relevant for an alternative fast-first ordering.

Lemma 25 (Fourier symbol of Method B)**.** *With $G(s)$ the optimal-step NIDC factor (10) of §1, the macro amplification factor for the quadratic extrapolant is*

$$M_B(s;\theta) = G(s)^{n_r}\left[\tfrac{(\theta+1)(\theta+2)}{2} - \tfrac{\theta(\theta+2)}{G(s)} + \tfrac{\theta(\theta+1)}{2\,G(s)^2}\right].$$

At the grid mode, the diffusion-only part of G equals $1/3$ — the value used in the bracket evaluation below; the full factor there is $\frac{1+a}{3} + \frac{a^2}{2}$ with $a = \eta/6$.

- **[Open]** Show $|M_B(s;\theta)| \leq e^{\alpha H}$ uniformly in $s \in [0,1]$ and $\theta \in [0,1)$ for the admitted n. At the grid mode this reduces to $3^{-n}(1+2\theta^2)$, so check the lower bound on n (i.e. on r_c) this implies.
- Do this numerically first and only then try to prove it.

4.4 Comparison of the two closures

- Table to include: stencil width (δ_x^4 five-point for A; none for B), history storage (none for A; $q+1$ levels of the fast species for B), required degree ($q \geq p$ for A, $q \geq p+1$ for B), stability argument (reduces to Theorem 19 for A; needs the macro-symbol analysis for B), behaviour as $\theta \to 1$.
- Recommendation to draft: A as the default (stability is inherited, no history); B as an alternative that avoids the five-point stencil in the final step but needs more history and a separate stability check.

4.5 Numerical validation

- Mirror Table 1 on a two-species linear system with $D_1/D_2 \in \{5, 10\}$, chosen so that $r > 4$, T and boundary conditions as in the scalar test, error against the exact cosine-series solution.
- Rows to include:
  - A with $q = 1$ (expected order 4).
  - A with $q = 0$, a deliberately under-resolved interpolant (expected order 2): negative control for Theorem 21.
  - B with $q = 1$ (expected order 2) and $q = 2$ (expected order 4): negative control and positive result for Theorem 24.
  - Forward Euler subcycled with the same step structure, as baseline.
- Include non-integer and near-integer r (e.g. $\theta \approx 0.1$ and 0.9) to exercise the remainder.
- Also scan $r \in \{3, 3.5, 3.9\}$ in Regime 1 against Regime 2 to justify r_c.

4.6 Open items

- Coupled (off-diagonal) stability for the subcycled-and-interpolated update; only the scalar building block is covered so far.
- Cross-term correction with $\partial_t u^{(2)}$ inside the sub-steps (see above).
- More than two species: define the ordering of the ratios and whether to nest the subcycles.
- $p \geq 2$: interpolation degree $q \geq p$ and extrapolation degree $q \geq p + 1$ are predicted by the same counting, but untested.

A The p-level tower

The general correction is generated by powers of the operator, with the sign of Remark 3:

$$C_j^{(\le p)} \;=\; +\sum_{\ell=1}^{p} \frac{\Delta t_{\mathrm{opt}}^{\ell}}{(\ell+1)!}\big(D\mathcal{D}_2+\alpha\big)^{\ell} u_j^n, \tag{18}$$

where $\mathcal{D}_2$ is a spatial operator whose accuracy must be raised as ℓ grows. Spatial discretization error breaks the binomial symmetry of the time expansion, giving the stencil-scaling bound:

Proposition 26 (Stencil bound; sketch)**.** *Global order $2p+2$ requires corrections evaluated on at least a $(2p+3)$-point stencil: $p=1$ needs the 3-point main stencil plus a 5-point evaluation inside $C^{(1)}$ when full fourth-order spatial accuracy is demanded of the correction itself; $p=2$ needs a 7-point stencil to represent the uncancelled u_{xxxxxx} residual.*

Status: [O]for $p \ge 2$ numerics (no harness yet; the bound's derivation is a consistency argument in `OTS-NIDC.md` §6). The $p=1$ row is [V]via Table 1.

B Notation dictionary and reproducibility

B.1 Global notation dictionary

Steps. The optimal (Chu) step is $\Delta t_{\mathrm{opt}} = \Delta x^2/(6D)$ for diffusivity D; for species i, $\Delta t_{\mathrm{opt}}^{(i)} = \Delta x^2/(6D_i)$, also written τ_i (and simply τ when the scalar problem is meant). A forced or actual step is h; the shared two-species step of §3 is $h = \Delta t_{\mathrm{opt}}^{(2)}$. The macro step of §4 is $H = \Delta t_{\mathrm{opt}}^{(2)}$, decomposed as n_r full steps of $\Delta t_{\mathrm{opt}}^{(1)}$ plus a remainder $\rho = \theta\,\Delta t_{\mathrm{opt}}^{(1)}$. The local truncation error is always $\mathcal{T}$ (never τ), as in (2).

Dimensionless groups. The master variables are the *step ratio* $r = h/\tau = 6D\Delta t/\Delta x^2$ (legacy diffusion number $\mu = r/6$, used only in classical comparisons) and the *step-independent reaction strength* $\eta = \alpha\Delta x^2/D$. The per-step reaction exponent is $a = \alpha h = r\eta/6$ (so $a = \eta/6$ at the optimum). The mismatch parameter is $\varepsilon = h - \tau = (r-1)\tau$; for the remainder step of Method A, $\varepsilon_\rho = \rho - \Delta t_{\mathrm{opt}}^{(1)} = (\theta-1)\Delta t_{\mathrm{opt}}^{(1)}$. The subcycle count is $n_r = \lfloor r \rfloor$ and the remainder fraction is $\theta = r - n_r$.

Amplification factors. With $s = \sin^2(\kappa\Delta x/2)$: G_{FE} is the factor of the full (diffusion + reaction) uncorrected forward-Euler scheme, $G_{\mathrm{FE}} = 1 - \frac{2r}{3}s + \frac{r\eta}{6}$ (§1); G_{NIDC} is the factor of the corrected scheme with the $C^{(1)}$ coefficients derived at the optimum, $G_{\mathrm{NIDC}} = G_{\mathrm{FE}} - \frac{\eta}{9}s + \frac{\eta^2}{72}$ (§1); G_{mis} is the *diffusion-only* factor of the mismatch-corrected diffusion step, $G_{\mathrm{mis}} = 1 - \frac{2r}{3}s + \frac{2r(r-1)}{9}s^2$ (§3); $G_{\mathrm{NIDC},h}$ is the full factor of the forced-step corrected scheme, $G_{\mathrm{NIDC},h} = G_{\mathrm{FE}} + \frac{2r(r-1)}{9}s^2 - \frac{2ra}{3}s + \frac{a^2}{2}$ (§3, Remark 18). $E(s)$ is the exact semi-discrete one-step factor $e^{-\frac{2r}{3}s+a}$; E_q (§4) is unrelated — the degree-q time extrapolant.

Bridge identities. $r = 6\mu$; $a = \mu\eta = r\eta/6$; $\eta = 6a$ at $r = 1$; $\mu = \frac{1}{6}$ (equivalently $r = 1$) at the optimal step. At $r = 1$ the factors agree: $G_{\mathrm{NIDC},h} = G_{\mathrm{NIDC}}$ and the diffusion part of G_{NIDC} equals G_{mis}. Off the optimum, G_{NIDC} (frozen $C^{(1)}$ coefficients) and $G_{\mathrm{NIDC},h}$ (h-rescaled corrections plus mismatch) are different conventions; the paper's stability statements use the former in §1 and the latter in §3, and both satisfy $|G| \le e^a$ on their respective admitted ranges.

Indices and spaces. Nodes are j (subscript); species are i, with reaction matrix $R_{i\ell}$ (the scalar-analysis reaction strength α is the diagonal eigen-reaction of the moment hierarchy); spatial wavenumber is κ; time level is n, the final time level is n_f, the subcycle count is n_r, and macro-step counter is k (t^k). Grid size is N_x. The spatial domain of the companion Part II is Ω (not D, which is reserved for diffusivity); its Hilbert space is $H = L^2(\Omega)$; the Mercer eigenvalues of the covariance are ν_p (not μ_p, avoiding the diffusion-number collision). The noise channel index is k, the covariance mode index p, and the Fourier mode index m — k and m are local to the companion Part II and never appear there as steps or nodes.

B.2 Reproducibility

The convergence harness of Table 1 is `code/implementation/convergence_test.py` in the repository, with results logged in `code/implementation/results.md` (2026-07). The stability claims of §1 and §3 were cross-checked by independent re-derivation of every amplification factor symbolically from the update formulas and by dense numerical sweeps over (r, η, s) — zero violations of the stated bounds on the admitted ranges; the check scripts live outside the repository tree and should be re-homed under `code/verification/` in the next release.

C Open items

- **Coupled discrete stability**: the off-diagonal reaction coupling and its interpolated subcycled variant have no independent stability proof; only the scalar building blocks of §1 and §3 are established.
- **Cross-term correction with $\partial_t u^{(2)}$** inside the sub-steps of §4 (supply via finite difference or Hermite derivative; check it does not lower the order).
- **More than two species**: define the ordering of the ratios and whether to nest the subcycles.
- $p \geq 2$: interpolation degree $q \geq p$ and extrapolation degree $q \geq p + 1$ are predicted by the same counting, but untested.

References

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